A CUBIC CRITERION FOR CONTINUOUS ALGEBRAIC BERNOULLI CONVOLUTIONS

Abstract. We extend the work of Kogler [Kog26] on continuous densities of algebraic Bernoulli convolutions, reducing the exponent in their criterion from 100 to 3.

1. The theorem and the route to continuity

We extend the work of Kogler [Kog26, Theorem 1.2 and equation (1.2)] by replacing the exponent 100 in their algebraic continuity criterion with 3.

Let ν_λ be the law of

$$X_\lambda = \sum_{n=0}^{\infty} \varepsilon_n \lambda^n, \qquad \mathbb{P}(\varepsilon_n = 1) = \mathbb{P}(\varepsilon_n = -1) = \tfrac{1}{2},$$

where the signs are independent. The Mahler measure is that of the primitive minimal polynomial with positive leading coefficient. All logarithms are natural except $\log_2$.

Theorem 1.1. *If $\lambda \in (1/2, 1)$ is algebraic of degree d_λ, has Mahler measure M_λ, and satisfies*

$$\lambda \geq 1 - \left(\frac{\min\{1, \log M_\lambda\}}{1 + d_\lambda + \log M_\lambda} \right)^3, \tag{1.1}$$

then ν_{λ^2} has an $L^2(\mathbb{R})$ density and ν_λ has a continuous, compactly supported density on $\mathbb{R}$.

Corollary 1.2 (Rational parameters)**.** *Let $\lambda = 1 - p/q \in (1/2, 1)$ be in lowest terms. If*

$$p \leq \frac{q}{(2 + \log q)^3}, \tag{1.2}$$

then ν_λ has a density in $C_c(\mathbb{R})$. Moreover, $\nu_{1-1/n}$ has a continuous density for every integer

$$n \geq 588.$$

Proof. The primitive minimal polynomial of λ is $qx - (q - p)$, so $d_\lambda = 1$ and $M_\lambda = q \geq 3$. Since $\log q > 1$, condition (1.1) becomes

$$\lambda \geq 1 - \frac{1}{(2 + \log q)^3},$$

which is equivalent to (1.2). Thus Theorem 1.1 applies.

For $p = 1$ and $q = n$, consider $F(x) = x/(2 + \log x)^3$. We have

$$F'(x) = \frac{\log x - 1}{(2 + \log x)^4} > 0 \qquad (x \geq 3),$$

Date: 14 September 2026.

and the endpoint bounds are

$$(2+\log 587)^3 > 587.43 > 587, \qquad (2+\log 588)^3 < 587.80 < 588.$$

Consequently the cubic condition holds in this family exactly for the integers $n \geq 588$, proving the assertion. □

Why work with the square parameter? Separate the even and odd digits. Independence gives

$$\nu_\lambda = \nu_{\lambda^2} * (D_\lambda)_* \nu_{\lambda^2}, \qquad D_\lambda(x) = \lambda x. \tag{1.3}$$

The convolution of two L^2 functions is continuous: write its increments as an L^2 inner product and use continuity of translations in L^2. Thus the task is to prove square-integrability for $\theta = \lambda^2$.

What the proof has to overcome. Different sign words can give the same sum. We therefore count the entropy of the distinct sums, rather than the number of words. Arithmetic gives both a lower bound for this entropy and a lower bound for the distance between distinct sums.

Entropy means that some uncertainty remains. We locate that uncertainty at distances within a factor of sixteen of one another. Such alternatives produce Fourier cancellation after a suitable shift of the digit block. Blocks using disjoint digits are independent, so their cancellation multiplies. Repeating the construction at many block lengths makes the cost of removing a Fourier smoothing summable.

Finally, the resulting density is close to a Gaussian in L^1. Self-similarity turns this coarse approximation into an L^2 bound: an exceptional concentration would have to reproduce itself on smaller cylinders. A short stopping argument rules this out.

The analytic statement we use. For a parameter $\theta \in (1/2, 1)$ let $X_n = \sum_{j=0}^{n-1} \varepsilon_j \theta^j$, with equal atoms combined, and let Δ_n be the distance between its closest distinct atoms. We measure discrete entropy H in bits and write η_v for the centred Gaussian density of variance v.

Proposition 1.3 (Working regularity criterion)**.** *Suppose* $0 < \rho \leq 1$*,* $L \geq 3$*, and for every* $n \geq 1$*,*

$$H(X_n) \geq \rho n, \qquad \Delta_n \geq 2^{-Ln}, \qquad \log_2\left(\frac{2}{1-\theta} + 2\right) \leq L + 2. \tag{1.4}$$

Put

$$\chi = -\log\theta, \qquad J = L/\rho, \qquad t = \log(150J). \tag{1.5}$$

If

$$\chi \leq \frac{1}{250}, \qquad \chi t^2 \leq \frac{\rho}{2}, \tag{1.6}$$

then $\nu_\theta = f_\theta\, dx$ *with* $f_\theta \in L^2(\mathbb{R})$*, and*

$$\|f_\theta - \eta_{1/(1-\theta^2)}\|_1 < 2\chi. \tag{1.7}$$

Here ρ is the entropy retained per digit, while L measures how finely one might need to look to distinguish sums. Their ratio J sets the spacing of the block lengths. The condition on χt^2 ensures that the cancellation gained at each length outweighs this spacing. We first check that the cubic hypothesis supplies the proposition's inputs. Sections 3–6 then prove the proposition.

2. How the cubic hypothesis supplies the inputs

For an auxiliary algebraic parameter $\theta \in (1/2, 1)$ set

$$X_n = \sum_{j=0}^{n-1} \varepsilon_j \theta^j, \quad d_\theta = [\mathbb{Q}(\theta) : \mathbb{Q}], \quad m_\theta = \log M_\theta, \quad D_\theta = 1 + d_\theta + m_\theta.$$

Equal atoms are combined before entropy is computed. Let H_e denote entropy in natural logarithms in the next lemma.

Lemma 2.1. *The minimum distance Δ_n between distinct atoms of X_n obeys*

$$\Delta_n > e^{-D_\theta n}, \qquad \log \frac{1}{1-\theta} \le m_\theta + (d_\theta - 1)\log 2.$$

Moreover $H_e(X_n) \ge hn$ for every n, where one may choose

$$h = \begin{cases} (3/10)\min\{1, m_\theta\}, & m_\theta < 2, \\ 1/2, & m_\theta \ge 2. \end{cases} \tag{2.1}$$

If θ is not an algebraic unit, $h = \log 2$ is also valid.

Proof. A difference of sign sums is $2P(\theta)$ for a polynomial P with coefficients in $\{-1, 0, 1\}$ and degree $k < n$. If $P(\theta) \neq 0$, the nonzero integral resultant with the minimal polynomial gives

$$|P(\theta)| \ge (k+1)^{1-d_\theta} M_\theta^{-k}.$$

Indeed bound $|P(z)|$ at each other conjugate by $(k+1)\max(1, |z|)^k$. This proves the separation bound, using $\log n \le n$. Evaluation of the minimal polynomial at one gives

$$1 \le |P_\theta(1)| \le 2^{d_\theta - 1} M_\theta (1 - \theta),$$

which proves the second assertion.

The entropy bound follows from the Gaussian comparison of Breuillard–Varjú [BV20, Proposition 13]. Its short derivation, including both constants in (2.1), is given in Appendix A.1. □

For the parameter λ in Theorem 1.1, we now use the abbreviations

$$\begin{gathered} d = d_\lambda, \quad m = \log M_\lambda, \quad s = \min\{1, m\}, \\ D = 1 + d + m, \qquad B = B_\lambda = D/s, \qquad g = 1 - \lambda. \end{gathered} \tag{2.2}$$

Here $m > 0$ and $B \ge 3$. For $\theta = \lambda^2$, the degree and Mahler measure satisfy

$$d_\theta \le d, \qquad m_\theta = \frac{2d_\theta}{d} m, \qquad m \le m_\theta \le 2m, \qquad D_\theta \le 2D. \tag{2.3}$$

Here is a direct polynomial proof. Let P and Q be the primitive minimal polynomials of λ and λ^2, with positive leading coefficients. If P is even, then $P(x) = Q(x^2)$. Otherwise

$$Q(x^2) = (-1)^d P(x) P(-x).$$

Indeed, in the second case the conjugates of λ have distinct squares, and Gauss' lemma makes the right side primitive. Taking degrees and Mahler measures proves (2.3). In particular, the h above is always at least $3s/10$.

Checking the cubic condition. Assume $gB^3 \le 1$. Evaluation of the minimal polynomial at one gives

$$3\log B \le \log(1/g) \le m + (d-1)\log 2 \le D - 2 \le B - 2. \tag{2.4}$$

This observation does two jobs. It excludes the small values of B where a uniform estimate might fail, and it prevents the entropy factor $s = D/B$ from being too small.

The concave function $3\log B - B + 2$ is positive at both 3 and 8. Consequently (2.4) forces $B > 8$, and hence

$$\chi = -\log(\lambda^2) = -2\log(1-g) \le \frac{2g}{1-g} < 2.004B^{-3} < \frac{1}{250}. \tag{2.5}$$

In particular $\theta = \lambda^2 > 1/2$. For this θ, choose

$$L = D_\theta/\log 2, \qquad \rho = h/\log 2, \qquad J = D_\theta/h, \tag{2.6}$$

where h is given by Lemma 2.1. Then $0 < \rho \le 1$ and $L \ge 3$: for degree at least two this is immediate, while a rational parameter has $m_\theta \ge \log 2$. The separation and entropy conditions in (1.4) follow from the lemma. The remaining condition follows from

$$\frac{2}{1-\theta} + 2 \le \frac{4}{1-\theta}, \qquad \log\frac{1}{1-\theta} \le D_\theta.$$

It remains to check just $\chi t^2 \le \rho/2$.

Completion of Theorem 1.1, assuming Proposition 1.3. There are two short cases, according to which entropy bound is useful.

If $m_\theta \ge 2$, take $h = 1/2$. Then $\rho = 1/(2\log 2)$ and $J = 2D_\theta \le 4B$. Thus

$$\chi t^2 < 2.004B^{-3}[\log(600B)]^2 \le \frac{2.004[\log 4800]^2}{8^3} < 0.282 < \frac{1}{4\log 2} = \frac{\rho}{2}.$$

The expression decreases for $B \ge 8$.

If $m_\theta < 2$, then $m < 2$. Rewriting the middle bound in (2.4) gives

$$3\log B \le (\log 2)D + (1-\log 2)m - 2\log 2 < (\log 2)B + 2 - 4\log 2.$$

The difference between the left and right sides is concave and positive at 3 and 11, so now $B > 11$. Moreover

$$s = D/B \ge \frac{3\log B}{B}, \qquad \rho \ge \frac{3s}{10\log 2}, \qquad J \le \frac{20}{3}B.$$

Set $T_B = \log(1000B)$, so that $t \le T_B$. Then

$$\frac{gT_B^2}{s} \le \frac{T_B^2}{3B^2\log B} \le \frac{[\log 11000]^2}{363\log 11} < 0.099485 < \frac{1}{10}. \tag{2.7}$$

Again the expression decreases on the indicated range. Since $g < 1/512$,

$$\frac{\chi t^2}{\rho} \le \frac{20\log 2}{3(1-g)}\frac{gT_B^2}{s} < \frac{2\log 2}{3}\frac{512}{511} < 0.464 < \frac{1}{2}.$$

Thus Proposition 1.3 applies in both cases and gives $\nu_{\lambda^2} \in L^2$. Equation (1.3) proves continuity. The support is compact because the defining series is bounded. □

The rest of the paper proves the working criterion. The arithmetic parameters d, m, B will no longer be needed.

3. Finding entropy at comparable distances

For the rest of the proof assume (1.4)–(1.6). Fix

$$c_0 = \frac{4}{25}, \qquad a_0 = \frac{13}{40}, \qquad \kappa = \chi/\log 2, \qquad \varepsilon_0 = \frac{1}{10^5 Jt}. \tag{3.1}$$

We have $J \geq 3$, $t \geq \log 450 > 6$, $\chi/\rho < 1/70$ and $\varepsilon_0 < 10^{-5}$. The letter j below measures distance on a dyadic logarithmic scale; it need not be an integer.

A finite entropy budget. A conditional decomposition of a finite law is a representation obtained by first sampling a label Z, then sampling from a conditional law π_Z. Averaging the conditional laws recovers the original law. Let $C_{a,16}(X_n)$ be the largest expected conditional entropy when each conditional law has all its distinct support distances in $[a, 16a)$. Such a law has at most sixteen points.

The elementary decomposition proved in Appendix A.2 gives

$$\int_{c_0\rho n}^{Ln+4} C_{2^{-j},16}(X_n)\, dj \geq 4(1-c_0)\rho n - 4C_\theta, \qquad C_\theta \leq L+2. \tag{3.2}$$

Here is the meaning of this estimate. At the coarse resolution $2^{-c_0\rho n}$, recording the position costs at most $c_0\rho n + C_\theta$ bits. At the fine resolution 2^{-Ln} every distinct atom is resolved. Thus at least $(1-c_0)\rho n - C_\theta$ bits must be revealed between the two scales. A band spanning a factor of sixteen counts four dyadic scales, explaining the factor four.

From uncertainty to cancellation. A conditional law with one dominant atom gives little cancellation; a spread-out law gives more. The next inequality connects this distinction to entropy. Its quantity $A(\pi)$ is the cancellation that we will obtain in the next section.

Lemma 3.1. *For every probability vector π on at most sixteen points, define*

$$A(\pi) = 1 - \sqrt{\frac{2}{5} + \frac{3}{5}\sum_i \pi_i^2}.$$

For every $\alpha > 0$,

$$H(\pi) \leq \alpha A(\pi) + \log_2(1 + 15\, 2^{-\alpha a_0}). \tag{3.3}$$

In particular, with

$$\alpha = \frac{t}{a_0 \log 2}, \qquad \beta = \log_2\left(1 + \frac{1}{10J}\right) \leq \frac{1}{10J\log 2}, \tag{3.4}$$

we have $H(\pi) \leq \alpha A(\pi) + \beta$.

Proof. Write $\delta = 1 - \max_i \pi_i$ and $\gamma = 3/5$. We first show $A(\pi) \geq a_0\delta$. If $\delta \leq 1/2$, then $\sum \pi_i^2 \leq (1-\delta)^2 + \delta^2$. The function

$$F(\delta) = 1 - \sqrt{1 - 2\gamma\delta(1-\delta)}$$

is concave, since $F'' = \gamma(\gamma-2)/(1-2\gamma\delta(1-\delta))^{3/2} < 0$. Its chord on $[0, 1/2]$ gives $F(\delta) \geq 2(1-\sqrt{7/10})\delta \geq a_0\delta$. If $\delta \geq 1/2$, use $\sum \pi_i^2 \leq \max_i \pi_i = 1-\delta$ and

$$\frac{1-\sqrt{1-\gamma\delta}}{\delta} = \frac{\gamma}{1+\sqrt{1-\gamma\delta}} \geq 2(1-\sqrt{7/10}) > a_0.$$

Write $H_2(u) = -u\log_2 u - (1-u)\log_2(1-u)$ for binary entropy. Grouping the other atoms gives $H(\pi) \le H_2(\delta) + \delta\log_2 15$. The binary entropy variational identity yields

$$H(\pi) - \alpha A(\pi) \le H_2(\delta) + \delta\log_2 15 - \alpha a_0\delta \le \log_2(1 + 15\,2^{-\alpha a_0}),$$

as required. Since $2^{-\alpha a_0} = 1/(150J)$, the last assertion follows. □

Choose maximizing distance-band decompositions measurably in j and write $\delta_n(j) = \mathbb{E}A(\pi_{Z_j})$. Only finitely many band graphs occur for fixed n, so this choice presents no measurability issue. The conditional labels keep their original probabilities. The entropy budget and Lemma 3.1 give

$$\int_{c_0\rho n}^{Ln+4} \delta_n(j)\,dj \ge nW - \frac{4C_\theta + 4\beta}{\alpha}, \tag{3.5}$$

$$W = \frac{4(1-c_0)\rho - (L-c_0\rho)\beta}{\alpha} \ge \frac{a_0\rho\log 2}{t}\left(\frac{84}{25} - \frac{1}{10\log 2}\right). \tag{3.6}$$

The last step uses $\beta \le 1/(10J\log 2) = \rho/(10L\log 2)$. Also $0 \le \delta_n(j) \le d_* := 1 - \sqrt{7/16}$.

4. Cancellation from independent digit blocks

Shifting a block separates its phases. The transform of a sum of independent variables is a product. To make one factor smaller than one, we need alternatives whose phases differ. The next estimate says that geometric shifts of a distance in a fixed band cannot all have nearly equal phases.

Put $U = 1/(2\theta)$ and $K = \lceil \log 10000/\chi \rceil$. Use weights $\omega_k = (1-\theta)\theta^k$, $0 \le k < K$, and $\omega_K = \theta^K$.

Lemma 4.1. *For $\theta U \le y \le 16U$,*

$$\sum_{k=0}^{K} \omega_k(1 - \cos(2\pi\theta^k y)) \ge \frac{3}{5}. \tag{4.1}$$

Proof. The infinite geometric weights are the lengths of $[\theta^{k+1}, \theta^k]$. The right-endpoint quadrature error for $\int_0^1 \cos(2\pi y u)\,du$ is at most $\pi y(1-\theta)/(1+\theta)$: integrate the Lipschitz bound on each interval and sum its squared length. For $1/2 \le y \le 1$ this integral is nonpositive, and for $y \ge 1$ it is at most $1/(2\pi)$. Moving the remaining weight to K costs at most $2\theta^K \le 1/5000$. The left side of (4.1) is therefore at least

$$1 - \frac{1}{2\pi} - \frac{8\pi(1-\theta)}{\theta(1+\theta)} - \frac{1}{5000} > 0.6585 > \frac{3}{5}.$$

The bound is smallest at $\theta = e^{-1/70}$. □

For a conditional law with support distances in $[a, 16a)$ and target scale r, choose k_0 so that $\theta^{k_0} a/(2\pi r) \in [\theta U, U]$. The identity

$$\left|\sum_i \pi_i e^{isx_i}\right|^2 = 1 - 2\sum_{i<j} \pi_i\pi_j(1 - \cos(s(x_i - x_j)))$$

and Lemma 4.1, followed by Cauchy–Schwarz in the shift variable, give

$$\sum_{k=0}^{K} \omega_k \, \mathbb{E}_Z \bigl|\widehat{\pi_Z}(\theta^{k_0+k}/r)\bigr| \le 1 - \mathbb{E}_Z A(\pi_Z).$$

Thus some common shift $k_0 + k$, $0 \le k \le K$, satisfies

$$\mathbb{E}|\widehat{\mu}_Z(1/r)| \le 1 - \mathbb{E}A(\pi_Z).$$

Here μ_Z is the conditional law after that fixed shift. The shift is chosen before the label is sampled. After translation, each shifted conditional support lies in $[0, 52r]$, because its diameter is at most $32\pi Ur = 16\pi r/\theta < 52r$. Thus on $\bigl||\xi| - 1\bigr| \le \varepsilon_0$ the loss of attenuation at ξ/r is at most $52\varepsilon_0$.

Why several block lengths are needed. One block length supplies a fixed amount of cancellation. As the target frequency grows, we can use more lengths, while keeping all the digits disjoint. The following choices make this elementary packing explicit:

$$\text{block lengths: } n_0, n_0N, n_0N^2, \dots.$$

The number N separates their ranges of possible digit positions; the large starting length n_0 absorbs fixed endpoint losses. Define

$$\begin{gathered} N = \lceil 9J \rceil, \quad n_0 = \lceil 10^6 Jt/\chi \rceil, \quad R_0 = 4(L+1)n_0, \\ a_* = \frac{0.999}{\kappa}\left(W - \frac{0.001\rho}{t}\right). \end{gathered} \tag{4.2}$$

The cancellation budget has the following numerical margin:

$$0.999\left[a_0(\log 2)^2\left(\frac{84}{25} - \frac{1}{10\log 2}\right) - 0.001\log 2\right] > 0.50093 > \frac{1}{2}. \tag{4.3}$$

Thus $\chi t^2 \le \rho/2$ implies

$$a_* \ge \frac{0.50093\rho}{\chi t} > t, \qquad Ne^{-a_*} \le \frac{10J}{150J} = \frac{1}{15}. \tag{4.4}$$

The decisive inequality is $Ne^{-a_*} < 1$: the cost of passing to the next block length is N, while the extra cancellation is e^{-a_*}. Everything below uses the bound $Ne^{-a_*} \le 1/15$.

Lemma 4.2. *For $r = 2^{-R}$ and $R \ge R_0$, put*

$$T = 1 + \left\lfloor \frac{\log(R/R_0)}{\log N} \right\rfloor. \tag{4.5}$$

There are at most LT/χ disjoint digit blocks, grouped into T lengths, whose summed attenuations on the Fourier band are at least a_*T. Their conditional supports, after scaling by r^{-1} and translation, lie in $[0, 52]$. The blocks use only digit indices i satisfying $R - \kappa i > 2$.

Proof. For a length $n \ge n_0$, put $h = (n+K+2)\kappa$ and extend $b_n(j) = (\delta_n(j) - 52\varepsilon_0)_+$ by zero outside $[c_0\rho n, Ln + 4]$. The averaging identity

$$\frac{1}{h}\int_0^h \sum_{\ell\in\mathbb{Z}} b_n(v + \ell h)\, dv = \frac{1}{h}\int_{\mathbb{R}} b_n(j)\, dj$$

shows that some translated grid has summed band attenuation at least

$$\frac{n(W-52\varepsilon_0 L)-10L}{(n+K+2)\kappa}\geq a_*. \tag{4.6}$$

Here $(4C_\theta+4\beta)/\alpha+208\varepsilon_0<10L$, $52\varepsilon_0 L=0.00052\rho/t$, and $10L/n\leq 10^{-5}\rho\chi/t<10^{-6}\rho/t$. Also

$$(K+2)/n\leq(\log 10000+3\chi)/(10^6 Jt)<10^{-6}.$$

These inequalities pay for the padding and the factor 0.999 in (4.6). Retain only grid points with positive b_n.

For $a=2^{-j}$, the first shift is $k_0=\lceil(R-j-\log_2(2\pi U))/\kappa\rceil$. Grid neighbours therefore have k_0 differing by exactly $n+K+2$, so their blocks remain disjoint after further shifts in $[0,K]$. In coordinate $R-\kappa k$ each block lies in

$$[(c_0\rho-\kappa)n-30,\ Ln+34].$$

Use the lengths $n_0, n_0N,\dots,n_0N^{T-1}$. Successive windows are disjoint because

$$N(c_0\rho-\kappa)\geq 9L(c_0-\kappa/\rho)>1.25L.$$

Indeed $0.25Ln_0>64$, which also absorbs the fixed padding. The largest length is at most $R/[4(L+1)]$, so all block starts are nonnegative. Moreover $(c_0\rho-\kappa)n_0-30>2$. The grid uses at most

$$\frac{(L-c_0\rho)n+4}{(n+K+2)\kappa}+1\leq L/\chi$$

blocks per length. These observations prove the assertions. The intervals and shifts are fixed before the conditional labels are sampled, independently for the disjoint blocks. □

5. Removing the smoothing

To obtain a density, we compare smoothings at successively finer scales. The many independent blocks make their differences summable. We also reserve a tail of unused digits: it supplies a very small initial factor, making the total error small enough for the later L^2 argument.

5.1. **A tail with small angles.**

Only the elementary inequality

$$0<\cos u\leq e^{-u^2/2}\qquad(|u|<\pi/2)$$

is needed. It follows by integrating $\tan u\geq u$ for $u\geq 0$.

For $r\leq\chi$, reserve the digits starting at an index ℓ such that

$$\frac{\sqrt{2}\,r}{1-\varepsilon_0}\leq\theta^\ell\leq\frac{\sqrt{2}\,r}{\theta(1-\varepsilon_0)}.$$

Such an index exists since the upper endpoint is less than one. On the band $\big||\xi|-1\big|\leq\varepsilon_0$, its first angle $A=|\xi|\theta^\ell/r$ lies in $[\sqrt{2},1.43]$. Consequently the tail product satisfies

$$|P_{\mathrm{tail}}(\xi)|=\prod_{j\geq 0}\cos(A\theta^j)\leq\exp\left(-\frac{A^2}{2(1-\theta^2)}\right)\leq E_*:=e^{-1/(2\chi)}. \tag{5.1}$$

Here $1-\theta^2\leq 2\chi$. The tail starts at $R-\kappa\ell<2$, so it is disjoint from the blocks in Lemma 4.2.

The sum of its coefficients after division by r is less than $2/\chi$. Each cosine is at least $\cos(1.43) > 1/8$, so omitting a factor costs at most eight. Termwise differentiation, justified by the summable coefficients, therefore gives

$$|P_{\mathrm{tail}}^{(j)}(\xi)| \le (16/\chi)^j E_*, \qquad j = 0, 1, 2. \tag{5.2}$$

5.2. **Localized Fourier inversion.** Choose an even $\widehat{\phi}$, equal to one on $[-1+\varepsilon_0, 1-\varepsilon_0]$ and zero outside $[-1-\varepsilon_0, 1+\varepsilon_0]$. On each transition use $1 - 10u^3 + 15u^4 - 6u^5$, $0 \le u \le 1$. Set $\widehat{\psi}(\xi) = -\xi\widehat{\phi}'(\xi)$ and $\phi_r(x) = r^{-1}\phi(x/r)$, similarly for ψ_r. With the convention $\widehat{g}(\xi) = \int e^{-i\xi x} g(x)\,dx$, we have

$$\|\widehat{\psi}\|_1 \le 2.1, \quad \|\widehat{\psi'}\|_1 \le 7/\varepsilon_0, \quad \|\widehat{\psi''}\|_1 \le 32/\varepsilon_0^2, \qquad \|\check{G}\|_1 \le \tfrac{1}{2}(\|G\|_1 + \|G''\|_1). \tag{5.3}$$

The quintic has vanishing first two derivatives at its endpoints, so $\widehat{\psi} \in W^{2,1}$; the kernels are integrable. The last inequality follows by applying Fourier inversion to $(1+x^2)\check{G}$ and integrating $(1+x^2)^{-1}$.

Condition on the graph labels and translate each conditional support to $[0, 52]$. For a retained block let b_i be its guaranteed attenuation, so $0 \le b_i \le d_*$ and $\mathbb{E}|P_i| \le 1 - b_i$ throughout the band. Its translated support gives $|P_i^{(j)}| \le 52^j$. In the product rule, differentiating a factor therefore costs at most $52/(1-b_i) \le 52/(1-d_*) < 80$ per derivative. Independence and the block count give the expected derivative bounds

$$\mathbb{E}\left|\partial_\xi^j \prod_i P_i(\xi)\right| \le (80LT/\chi)^j e^{-a_* T}, \qquad 0 \le j \le 2.$$

Together with (5.2), the product of the graph blocks and reserved tail has derivative bounds $D^j E_* e^{-a_* T}$, where $D = 100J(T+1)/\chi$. Translations depending on the labels preserve the L^1 norm. The remaining digits give an independent convolution by a probability measure, which can only decrease that norm. Therefore (5.3) implies

$$\|\nu_\theta * \psi_r\|_1 \le B_*(T+1)^2 e^{-a_* T}, \quad B_* = 2 \cdot 10^{11} J^2 t^2 \left(1 + \frac{0.001}{t\chi}\right)^2 e^{-1/(2\chi)}. \tag{5.4}$$

To see the prefactor directly, the inversion estimate is bounded by $20\varepsilon_0^{-2}(1 + \varepsilon_0 D)^2 E_* e^{-a_* T}$. For $\log_2(1/\chi) \le R < R_0$, use the reserved tail alone; the same bound holds with $T = 0$.

5.3. **Summing the changes of scale.** The intervals in (4.5) have length at most $R_0 N^T$ and $\sum_{T \ge 0} (T+1)^2/15^T < 2$. Also $R_0 \le 6 \cdot 10^6 J^2 t/\chi$. Thus

$$\int_0^\chi \|\nu_\theta * \psi_r\|_1 \frac{dr}{r} \le 2B_* R_0, \tag{5.5}$$

$$\frac{2B_* R_0}{\chi} \le 2.4 \cdot 10^{18} J^4 t^3 \chi^{-2} \left(1 + \frac{0.001}{t\chi}\right)^2 e^{-1/(2\chi)}. \tag{5.6}$$

Use $J = e^t/150$, $6 < t \le (2\chi)^{-1/2}$. The last expression is at most

$$\frac{2.4 \cdot 10^{18}}{150^4\, 2^{3/2}}\, \chi^{-7/2} \left(1 + \frac{1}{6000\chi}\right)^2 \exp\left(-\frac{1}{2\chi} + \frac{2\sqrt{2}}{\sqrt{\chi}}\right) < \frac{1}{100}. \tag{5.7}$$

This envelope increases for $0 < \chi \le 1/250$: its logarithmic derivative with respect to $\log\chi$ is at least $1/(2\chi) - \sqrt{2}/\sqrt{\chi} - 11/2 > 0$. At $\chi = 1/250$ it is less than $7 \cdot 10^{-18}$.

We now use the estimate for its intended purpose. The identity

$$\nu * \phi_a - \nu * \phi_b = \int_a^b \nu * \psi_r \, \frac{dr}{r} \qquad (0 < a < b)$$

shows that the smoothings converge in L^1 as $r \downarrow 0$. Since $\phi \in L^1$ and $\int \phi = \widehat{\phi}(0) = 1$, they also converge weakly to ν_θ. Thus $\nu_\theta = f_\theta \, dx$ and

$$\|f_\theta - \nu_\theta * \phi_\chi\|_1 < \chi/100. \tag{5.8}$$

6. Self-similarity upgrades the density to L^2

Only a coarse Gaussian approximation is needed. The variance of ν_θ is $1/(1-\theta^2)$. Appendix B proves, by expanding $\log \cos u$ only through its quadratic term, that

$$\|\nu_\theta * \phi_\chi - \eta_{1/(1-\theta^2)}\|_1 < \frac{9}{5}\chi. \tag{6.1}$$

Together with (5.8) this proves (1.7). The error need only be $O(\chi)$. After normalizing the support to $[0,1]$, the Gaussian height is $O(\chi^{-1/2})$, so their product tends to zero. The following lemma explains why that product is the relevant quantity.

Lemma 6.1 (Stopping at the first concentration)**.** *Normalize ν_θ to a probability density F on $[0,1]$, and write $\sigma = F\,dx$. Suppose that for every measurable set E,*

$$\sigma(E) \le K|E| + e, \qquad K \ge 1, \quad e > 0.$$

If $12Ke < \theta$, then $F \in L^2$.

Proof. The maps of the normalized system are $x \mapsto \theta x$ and $x \mapsto \theta x + 1 - \theta$. A word w gives an interval $I_w = g_w[0,1]$ and, conditional on that word, the law $(g_w)_*\sigma$. Set

$$a = e/K, \qquad r = \theta a, \qquad D_0 = 3/r, \qquad \beta = 2e.$$

Initially let E be open and $|E| \le r$. Along each symbolic path, stop at the first interval for which $|E \cap I_w| > r|I_w|$. Failure of this inequality at its parent implies

$$|g_w^{-1}E \cap [0,1]| \le r/\theta = a.$$

Thus, even at a stopped word, the conditional probability of ending in E is at most $Ka + e = \beta$.

Every path ending in E eventually stops, since E is open and cylinder diameters tend to zero. Stopped words are disjoint symbolic events. If Ω is the union of their intervals, their total probability is at most $\sigma(\Omega)$. Consequently

$$\sigma(E) \le \beta\sigma(\Omega).$$

Choose disjoint stopped intervals greedily in decreasing order of length. Their triples cover Ω. Each selected interval contains more than an r-fraction of E, so $|\Omega| \le 3|E|/r = D_0|E|$.

Put $\omega(t) = \sup_{|E|\le t} \sigma(E)$. Outer approximation and absolute continuity give

$$\omega(t) \le \beta\omega(D_0 t), \qquad 0 < t \le r.$$

Iterating until $D_0^j t > r$ yields $\omega(t) \ll t^\alpha$, where $\alpha = \log(1/\beta)/\log D_0$. For $m(u) = |\{F > u\}|$, the inequality $u m(u) \le \omega(m(u))$ implies $m(u) \ll u^{-1/(1-\alpha)}$. Here $0 < \alpha < 1$, since $\beta D_0 = 6K/\theta > 1$. The layer-cake integral $\int F^2 = 2\int_0^\infty u\, m(u)\, du$ is finite if $\alpha > 1/2$. This is exactly

$$\beta^2 D_0 = 12Ke/\theta < 1. \qquad \square$$

Completion of Proposition 1.3. We already have an absolutely continuous law and $\|f_\theta - \eta_{1/(1-\theta^2)}\|_1 < 2\chi$. Two probability densities differ on every measurable set by at most half their L^1 distance. After normalizing the support to $[0,1]$, the comparison Gaussian has maximum

$$K_0 = \sqrt{\frac{2}{\pi}}\sqrt{\frac{1+\theta}{1-\theta}} \le \frac{6}{5\sqrt{\chi}} =: K.$$

For example, $1-e^{-\chi} \ge \chi - \chi^2/2$ gives $\chi K_0^2 \le 4/[\pi(1-\chi/2)] < 36/25$. The stopping lemma therefore applies with $e = \chi$, because

$$12Ke \le \frac{72}{5}\sqrt{\chi} \le \frac{72}{5\sqrt{250}} < 0.911 < e^{-1/250} \le \theta.$$

This proves $f_\theta \in L^2$ and completes the argument. □

Appendix A. The entropy inputs

A.1. **The algebraic entropy bound.**

Proof of the entropy estimate in Lemma 2.1. The sequence $H_e(X_n)$ is subadditive, hence it is at least n times its limiting entropy rate h_θ. A coincidence of sign words would make θ a root of a monic integer polynomial with constant term ± 1, after removing initial zero coefficients. Thus nonunits have no coincidences and $h_\theta = \log 2$.

For units, $\prod_{|\theta_i|<1} |\theta_i|^{-1} = M_\theta$, since the product of all conjugate moduli is one. Thus Breuillard–Varjú [BV20, Proposition 13] gives, in natural logarithms,

$$h_\theta \ge \sup_{u>0}\{I(uM_\theta) - I(u)\}, \qquad I(u) = \mathrm{Ent}(u\varepsilon + G) - \mathrm{Ent}(G), \tag{A.1}$$

where G is standard Gaussian, independent of the fair sign ε, and Ent denotes differential entropy. Gaussian differentiation gives

$$I'(u) = u\mathbb{E}\,\mathrm{Var}(\varepsilon \mid u\varepsilon + G), \qquad \mathbb{E}\,\mathrm{Var}(\varepsilon \mid u\varepsilon + G) = e^{-u^2/2}\mathbb{E}\,\mathrm{sech}(uG) \ge e^{-u^2}.$$

The identity is the scalar Gaussian information/MMSE formula [GS(V05, Theorem 1]; it also follows by differentiation of this two-Gaussian mixture. The inequality uses $\log\cosh x \le x^2/2$ and $\mathbb{E}e^{-u^2G^2/2} = 1/\sqrt{1+u^2} \ge e^{-u^2/2}$. Integrate from $u = e^{-m_\theta/2}$ to $u = e^{m_\theta/2}$ to obtain

$$h_\theta \ge \Psi(m_\theta), \qquad \Psi(v) = \tfrac{1}{2}(e^{-e^{-v}} - e^{-e^{v}}).$$

The function Ψ increases, is concave on $[0,\infty)$, vanishes at zero, and satisfies $\Psi(1) > 0.313$. Concavity follows from

$$2\Psi''(v) = ae^{-a}(a-1) + be^{-b}(1-b) \le 0, \qquad a = e^{-v}, \quad b = e^{v}.$$

It follows that $h_\theta \ge (3/10)\min\{1, m_\theta\}$.

When $m_\theta \ge 2$, take $u = 2/M_\theta$ in (A.1). The sign estimator for $2\varepsilon + G$ has error at most $\varphi(2)/2 < 1/32$, where φ is the standard Gaussian density. Binary entropy and the Gaussian maximum-entropy bound yield

$$I(2) \ge \log 2 - \frac{1+\log 32}{32}, \qquad I(2/M_\theta) \le 2e^{-4}.$$

Their difference exceeds 0.516, proving (2.1). This derivation does not rely on the numerical constant discussed after Theorem 5 of [BV20]. □

A.2. **The finite decomposition behind the entropy budget.** The purpose of this appendix is to justify (3.2). It uses only the ordering of finitely many points on the line. Graph entropy is convenient notation for the construction; the required duality is proved here. It is a special case of the entropy-splitting principle in [CKL$^+$90]. For a finite probability vector π on the vertices of a graph G, let P_G be the convex hull of the indicators of independent sets, and define

$$\mathsf{H}_G(\pi) = \min_{x \in P_G} \sum_i \pi_i \log_2(1/x_i).$$

Zero-mass vertices may be removed. Let N_a join pairs of points at distance less than a, and let F_a be its complement.

Lemma A.1. *For finite probability laws on the line,*

$$\mathsf{H}_{N_a} + \mathsf{H}_{F_a} = H(\pi).$$

If $C_{a,16}$ is the greatest expected conditional entropy in a mixture whose distinct conditional support points have distances in $[a, 16a)$, then

$$C_{a,16} \geq \mathsf{H}_{N_{16a}} - \mathsf{H}_{N_a}. \tag{A.2}$$

Every conditional support in this statement has at most 16 *points.*

Proof. Independent sampling of independent sets gives $\mathsf{H}_{G_1 \cup G_2} \leq \mathsf{H}_{G_1} + \mathsf{H}_{G_2}$ by intersecting the sets. If w_S are weights of an optimal independent-set representation for $\overline{G}$, with coverages x_i, choose a label by

$$\mathbb{P}(Z = S \mid X = i) = w_S \mathbf{1}_{i \in S}/x_i.$$

Its conditional supports are cliques of G. If $\widetilde{w}$ is the label marginal, then $I(X;Z) = \mathsf{H}_{\overline{G}} - D(\widetilde{w}\|w) \leq \mathsf{H}_{\overline{G}}$. Consequently their expected entropy is at least $H(\pi) - \mathsf{H}_{\overline{G}}$.

For completeness, near/far duality follows from the order on the line. Let x minimize H_{N_a} and put $r_i = \pi_i/x_i$. Differentiating towards any independent-set indicator gives $\sum_{i \in S} r_i \leq 1$. Order $i \prec j$ when $z_j - z_i \geq a$. Its chains are the independent sets of N_a. Let T_i be the largest weight of a predecessor chain. The intervals $[T_i, T_i + r_i) \subset [0,1]$ are disjoint for comparable vertices. Sampling a uniform point gives an antichain with coverages r_i, hence an independent set of F_a. This proves $\mathsf{H}_{F_a} \leq H(\pi) - \mathsf{H}_{N_a}$. Conversely, independent sets in complementary graphs intersect in at most one vertex; their coverage vectors satisfy $\sum_i x_i y_i \leq 1$. Gibbs' inequality gives the reverse bound.

The complement of the distance-band graph is $N_a \cup F_{16a}$. Combine the preceding facts to obtain (A.2). If a band clique has k ordered points, then $(k-1)a < 16a$, proving the cardinality assertion. The strict upper endpoint is relevant here. □

Put $F_n(j) = \mathsf{H}_{N_{2^{-j}}}(\mathcal{L}(X_n))$. This function is decreasing and vanishes for $j \geq Ln$. Quantization into cells of length $2^{-c_0 \rho n}$, together with Lemma A.1, gives

$$F_n(c_0\rho n) \geq (1 - c_0)\rho n - C_\theta, \qquad C_\theta \leq \log_2\left(\frac{2}{1-\theta} + 2\right) \leq L + 2.$$

Indeed the cells are independent sets of the far graph, and assigning their probabilities as weights bounds its graph entropy by the entropy of the cell label. The number of cells is at most $2^{c_0\rho n}(2/(1-\theta) + 2)$. For distance band $[a, 16a)$, telescoping gives

$$\int_{c_0\rho n}^{Ln+4} C_{2^{-j},16}(\mathcal{L}(X_n))\, dj \geq \int_{c_0\rho n - 4}^{c_0\rho n} F_n(j)\, dj \geq 4(1 - c_0)\rho n - 4C_\theta.$$

Appendix B. An elementary Gaussian estimate

We prove (6.1) after normalizing to variance one. Put

$$\tau = 1-\theta^2, \quad a_n = \sqrt{\tau}\,\theta^n, \quad \Phi(u) = \prod_{n\ge 0}\cos(a_n u), \quad G(u) = e^{-u^2/2}.$$

Then $\sum a_n^2 = 1$ and $\sum a_n^4 = \tanh\chi \le \chi$. The transform of the smoothed law is $\Phi(u)q(u)$, where $q(u) = \widehat{\phi}(\chi\sqrt{\tau}\,u)$. Weighted Plancherel gives

$$\|h\|_1^2 \le \frac{1}{2}\int_{\mathbb{R}} (|\widehat{h}(u)|^2 + |\widehat{h}'(u)|^2)\,du. \tag{B.1}$$

Indeed apply Cauchy–Schwarz with weight $(1+x^2)^{-1}$ and then Plancherel to h and xh.

Low frequencies: a quadratic approximation. For $0 \le v \le 1$, the positive Taylor coefficients of tangent show

$$0 \le \tan v - v \le \tfrac{2}{3}v^3, \qquad 0 \le -\log\cos v - v^2/2 \le v^4/6.$$

For the first bound divide by v^3; the resulting power series increases, and $\tan 1 - 1 < 2/3$. Integrating proves the second. For $|u| \le u_0 := \tau^{-1/2}$, the cutoff is one and hence

$$\Phi(u) = G(u)e^{-R(u)}, \qquad 0 \le R(u) \le \chi u^4/6, \qquad |R'(u)| \le 2\chi|u|^3/3.$$

Using $1 - e^{-R} \le R$ gives

$$|\Phi - G| \le \frac{\chi}{6}u^4 G, \qquad |\Phi' - G'| \le \frac{\chi}{6}(|u|^5 + 4|u|^3)G.$$

The low-frequency contribution to the square-root norm in (B.1) is therefore at most $C\chi$, where

$$C^2 = \frac{\Gamma(11/2) + 9\Gamma(9/2) + 16\Gamma(7/2)}{72}, \qquad C < 1.709 < \frac{7}{4}. \tag{B.2}$$

This uses only $\int_{\mathbb{R}} u^{2j}e^{-u^2}\,du = \Gamma(j+1/2)$.

High frequencies: the small digits suffice. For $|u| \ge u_0$, choose the tail whose first angle lies in $[1, 1/\theta] \subset [1, 1.01]$. On this tail $|\cos v| \le e^{-v^2/2}$, and the sum of the squared angles is at least $1/(2\chi)$. Consequently, with $E_0 = e^{-1/(4\chi)}$,

$$|\Phi(u)| \le E_0, \qquad |\Phi'(u)| \le \frac{3}{\sqrt{\chi}}E_0.$$

For the derivative, omitting one tail factor costs at most two, since $\cos(1.01) > 1/2$; also $\sum a_n = \sqrt{(1+\theta)/(1-\theta)} < 3/(2\sqrt{\chi})$. Factors before the chosen tail can be discarded in each differentiated term.

The cutoff has support in $[-S, S]$ with $S = 2\chi^{-3/2}$. For the quintic transition, direct integration gives $\int|\widehat{\phi}'|^2 = 10/(7\varepsilon_0)$. Rescaling therefore gives $\int|q'|^2 \le 6\chi^{3/2}/\varepsilon_0$. Thus the high-frequency contribution of Φq is at most

$$E_0\left[S(1+18/\chi) + 6\chi^{3/2}/\varepsilon_0\right]^{1/2} < \chi/100. \tag{B.3}$$

Here $\varepsilon_0^{-1} = 10^5 Jt$, and

$$J = e^t/150, \qquad t \le (2\chi)^{-1/2}, \qquad \varepsilon_0^{-1} \le \frac{10^5}{150\sqrt{2\chi}}e^{1/\sqrt{2\chi}}. \tag{B.4}$$

After division by χ and squaring, the bound in (B.3) is a sum of positive multiples of

$$\chi^{-7/2}e^{-1/(2\chi)}, \quad \chi^{-9/2}e^{-1/(2\chi)}, \quad \chi^{-1}e^{-1/(2\chi)+1/\sqrt{2\chi}}.$$

All increase on $(0, 1/250]$. Substitution of $\chi = 1/250$ gives a square-root ratio below 10^{-20}, proving the stated loose bound.

For the Gaussian itself, set $y = 1/(2\chi) \le u_0^2$. The estimates

$$\int_b^\infty e^{-u^2}\,du \le \frac{e^{-b^2}}{2b}, \qquad \int_b^\infty u^2 e^{-u^2}\,du \le \left(\frac{b}{2} + \frac{1}{4b}\right) e^{-b^2}$$

give the high-frequency square-root bound

$$\left[\left(\frac{\sqrt{y}}{2} + \frac{3}{4\sqrt{y}}\right) e^{-y}\right]^{1/2} < \chi/100. \tag{B.5}$$

After division by χ and squaring, the two terms are constant multiples of

$$\chi^{-5/2}e^{-1/(2\chi)} \quad \text{and} \quad \chi^{-3/2}e^{-1/(2\chi)}.$$

Both increase on $(0, 1/250]$. The endpoint square-root ratio is below 10^{-23}.

Finally, split the integrals in (B.1) into low and high frequencies, and apply the triangle inequality to their square-root norms. This does not differentiate a sharp cutoff. Equations (B.2), (B.3) and (B.5) yield

$$\|\nu_\theta * \phi_\chi - \eta_{1/(1-\theta^2)}\|_1 < \left(\frac{7}{4} + \frac{1}{100} + \frac{1}{100}\right)\chi < \frac{9}{5}\chi,$$

as required.

References

[BV20] E. Breuillard and P. P. Varjú, *Entropy of Bernoulli convolutions and uniform exponential growth for linear groups*, J. Anal. Math. **140** (2020), 443–481. `https://arxiv.org/abs/1510.04043`; Proposition 13.

[CKL+90] I. Csiszár, J. Körner, L. Lovász, K. Marton, and G. Simonyi, *Entropy splitting for antiblocking corners and perfect graphs*, Combinatorica **10** (1990), 27–40.

[GS(V05] D. Guo, S. Shamai (Shitz), and S. Verdú, *Mutual information and minimum mean-square error in Gaussian channels*, IEEE Trans. Inform. Theory **51** (2005), 1261–1282.

[Kog26] C. Kogler, *On L^p density of self similar measures and Bernoulli convolutions*, 2026. Manuscript, September 2026.